Determining the Future Value of an Investment

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Principles of Finance

Introduction to Valuation: The Time Value of Money. – Chapter 2

Understanding how to calculate the future value of an investment is a crucial skill in finance. This presentation will guide you through the principles and formulas to determine the potential growth of your investments over time.

Learning Objectives

- Determine the future value of an investment made today.

Future Value (FV)

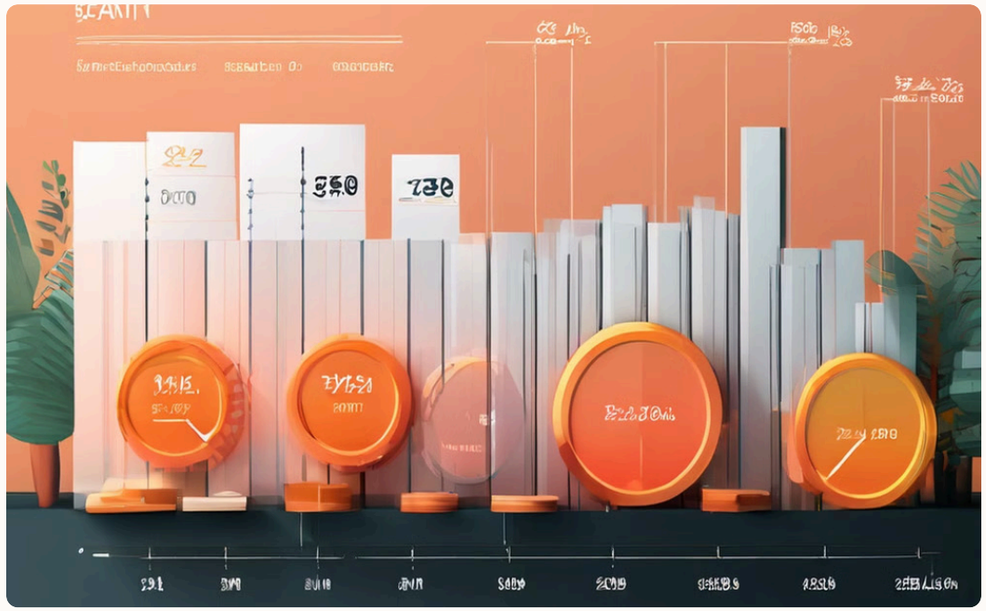
1 **Amount an investment will grow to over time.**

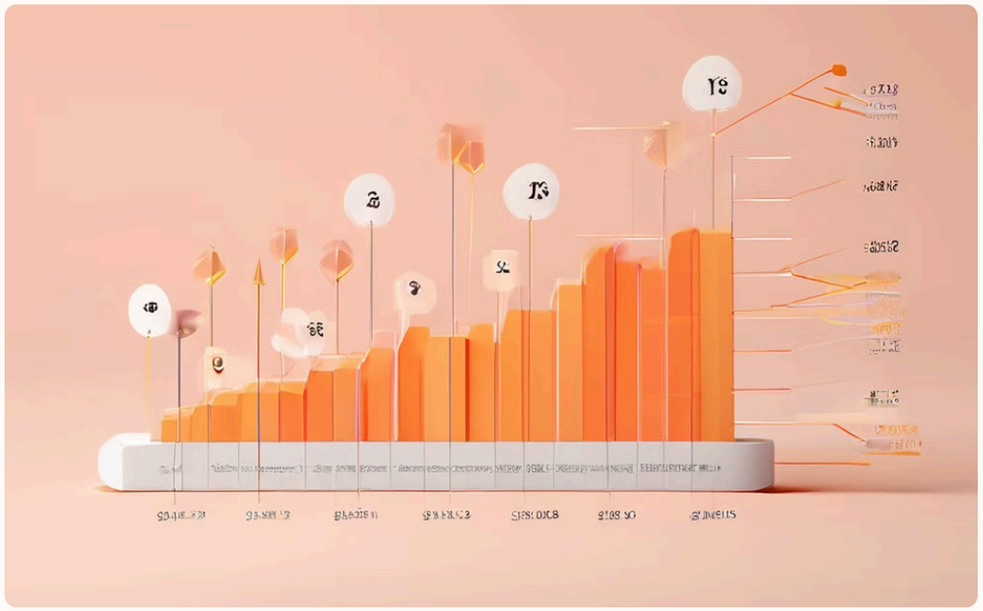
The future value represents the final sum your investment will reach after accounting for interest and time.

2 **Depends on initial investment (PV), interest rate (r), and time (t).**

The future value is determined by the principal amount invested, the annual interest rate, and the number of years the investment is held.

Simple vs. Compounded Interest





Simple Interest

Simple interest is calculated only on the principal amount. It is a fixed percentage of the original investment that is earned each period.

Compounded Interest

Compounded interest is calculated on the principal amount plus any accumulated interest. It earns interest not only on the initial investment but also on the interest earned in previous periods.

For example, imagine a student invests 10,000 Saudi Riyal in a bank account that offers a simple interest rate of 5% per year. After one year, they will earn 500 Riyal in interest (10,000 * 0.05). However, if they choose a compounded interest account, they will earn interest not only on the initial investment but also on the accumulated interest. This means that in the second year, they will earn interest on the original 10,000 Riyal plus the 500 Riyal earned in the first year, resulting in higher earnings.

Throughout this course we will focus on compounded interest.

Single-Period Investment

Formula:

$$FV = PV \times (1 + r)$$

Example:

1,000 at 10% interest in 1 year = 1,100

Multi-Period Investment

Formula:

$$FV = PV \times (1 + r)^t$$

Example:

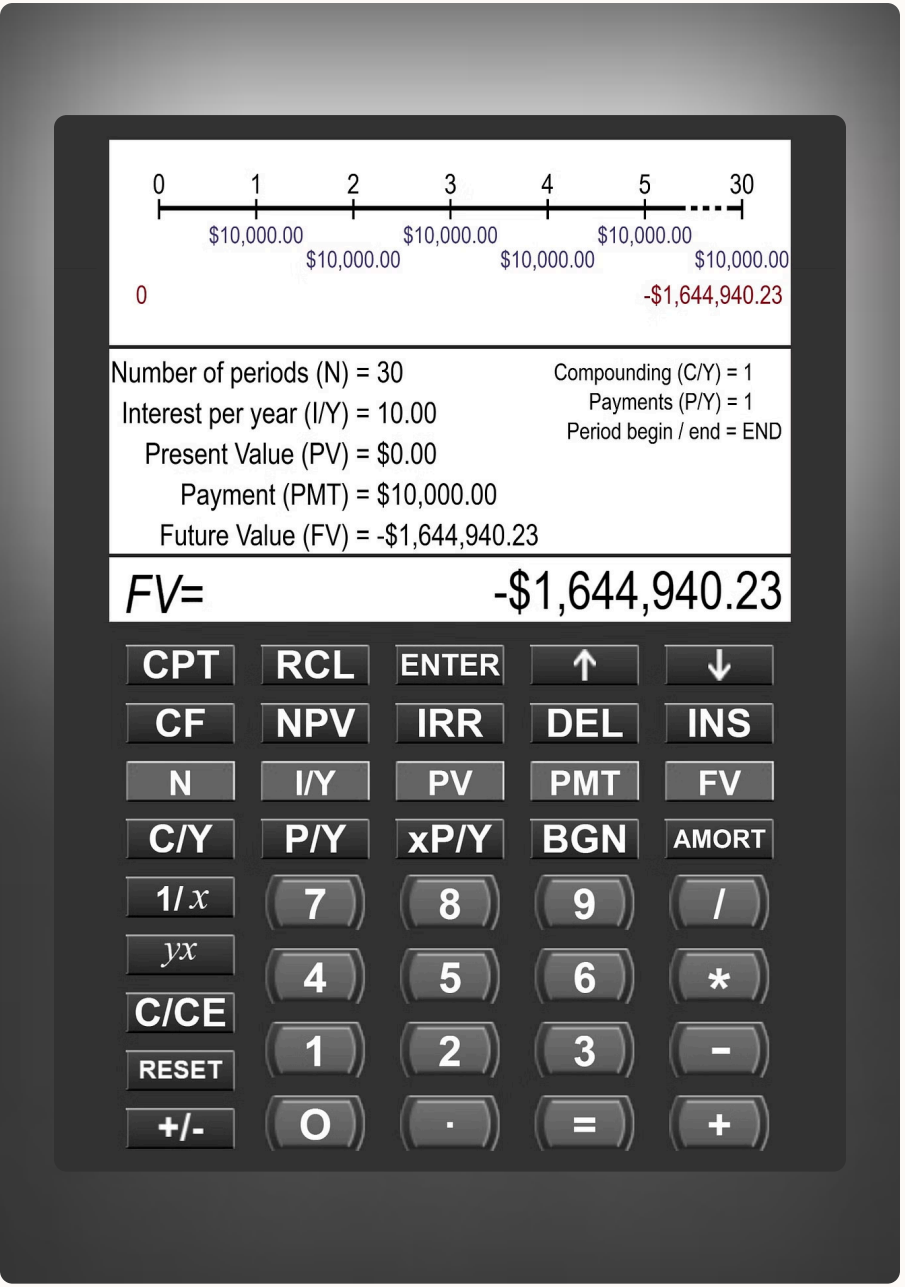
10,000 at 8% interest for 5 years = 14,693

Calculating Future Value in Excel

Excel provides a built-in function to easily determine the future value of an investment. Simply type the formula:

=FV(0.08, 5, 0, -10000)

This will display the future value of 14,693 after 5 years at an 8% interest rate, starting with an initial investment of $10,000.



Calculating Future Value Using a Financial Calculator

-10,000

Present Value (PV)

8%

Interest Rate (I/Y)

5

Years (N)

14,693

Future Value (FV)

The investment will grow to this amount after 5 years at an 8% interest rate.

Importance of Future Value

Evaluates investment opportunities.

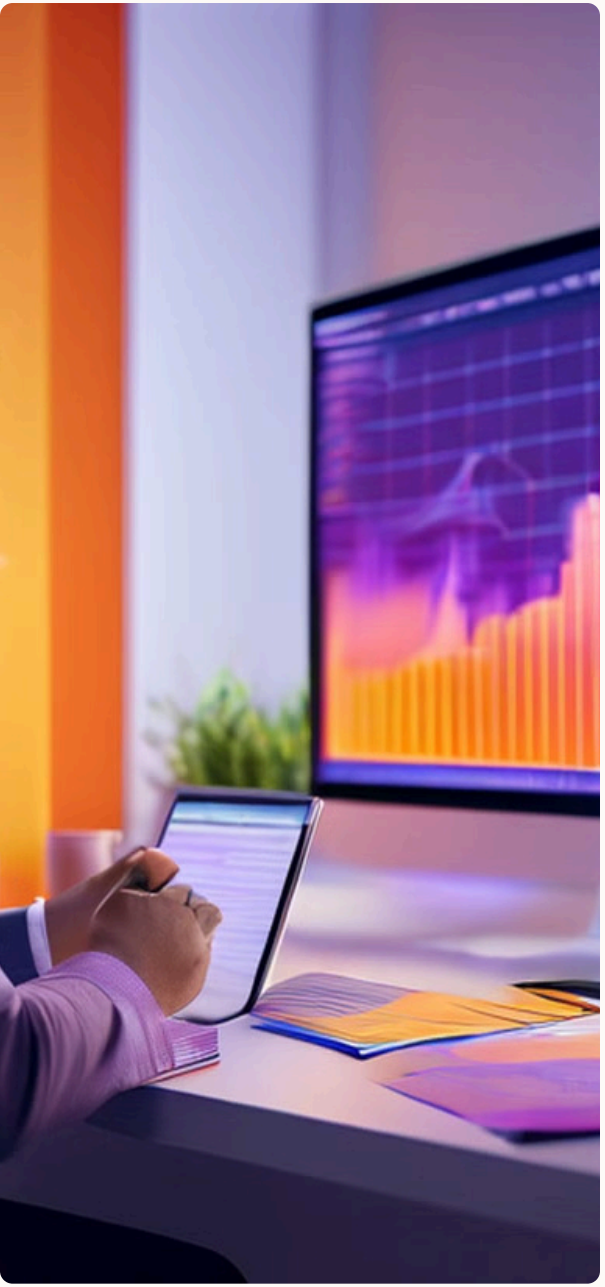
Calculating future value allows you to compare potential returns and make informed investment decisions.

Aids in savings and financial planning.

Understanding future value helps you set realistic savings goals and plan for major expenses or retirement.

Helps predict future worth of current investments.

Knowing the future value of your existing investments allows you to track their growth and make adjustments as needed.



Conclusion

Essential for financial decision-making.

The ability to calculate future value is fundamental to making sound financial choices and maximizing your investment returns.

Foundation for advanced financial concepts.

Understanding future value lays the groundwork for more complex financial principles, such as time value of money and investment analysis.

Key to making sound investment decisions.

By accurately projecting the future value of investments, you can make informed choices that align with your financial goals and risk tolerance.

Key Takeaways

Future value represents the growth of an investment over time.

It depends on the principal amount, interest rate, and time period.

Single-period and multi-period formulas are used for calculations.

Understanding future value is crucial for financial planning and decision-making.

Determining the Present Value of Future Cash Flows

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Determine the Present Value of Cash to be Received at a Future Date – Chapter 2

Understanding how to determine the present value of future cash flows is a fundamental concept in finance. It allows you to evaluate investment opportunities, compare different cash flows, and make informed financial decisions.

Learning Objectives

1 **Key Objective:**

Determine the present value of cash to be received at a future date.

Present Value Concept

1

Current value of a future amount of money.

2

Reflects the time value of money.

Single-Period Present Value

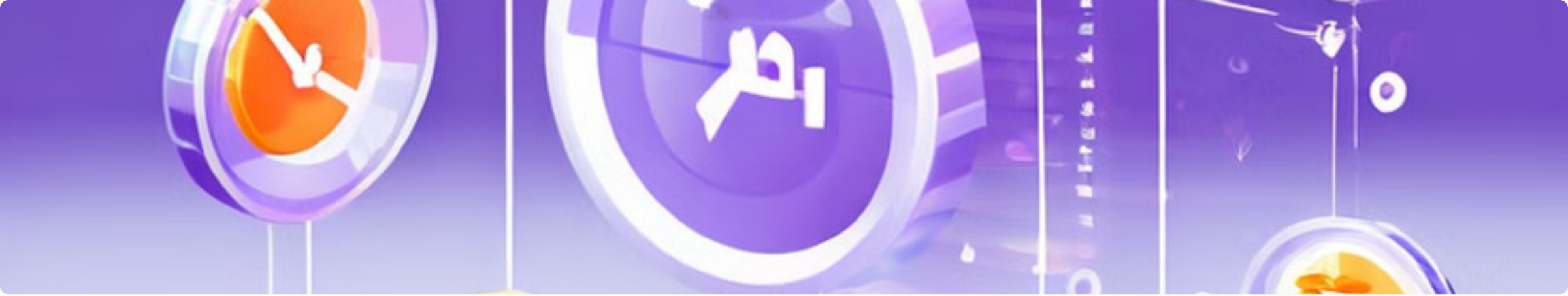
Formula

$$PV = FV/(1+r)$$

Example

1,000 to be received in 1 year at 10% discount rate

= 909.09



Multi-Period Present Value

Formula

$$PV = FV/(1+r)^t$$

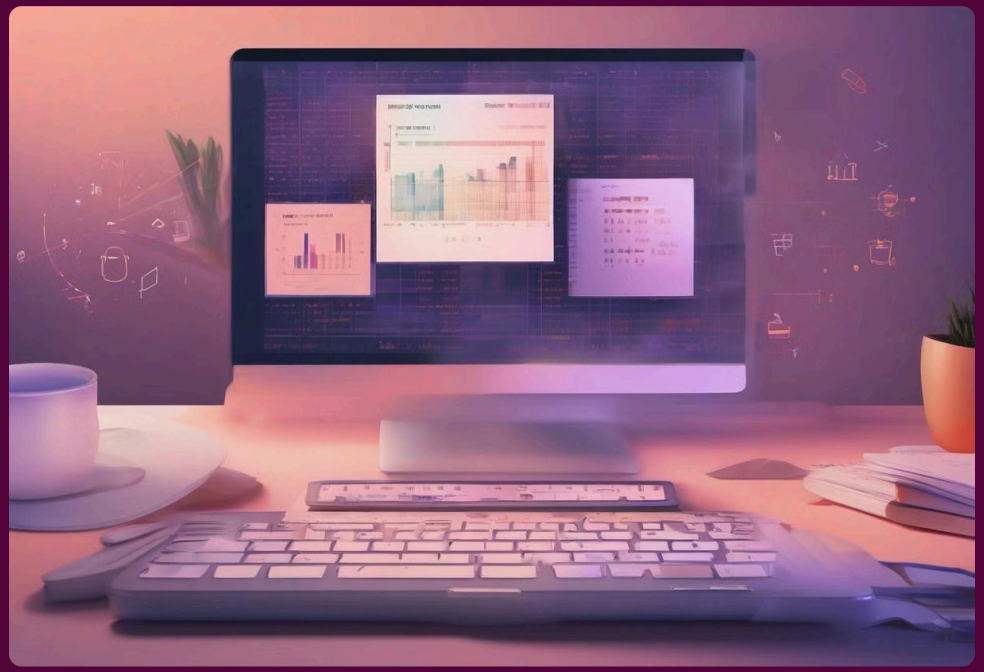
Example

5,000 to be received in 3 years at 7% discount rate

= 4,081.54

Using Excel for Present Value Calculation

1. Type `=PV(0.07, 3, 0, 5000)` and press Enter.
2. Excel will display the present value of 4,081.54.



Calculating Present Value Using a Financial Calculator

5000	7%
Future Value (FV)	Interest Rate (I/Y)
3	=4,081.54
Years (N)	Present Value (PV)

Perpetuity and Investment

A perpetuity is a stream of cash flows that continues forever. In Saudi real estate, this can be represented by long-term rental income. To calculate the present value of a perpetuity, we divide the annual cash flow by the discount rate.

$$PVofPerpetuiry = \frac{Cash\ flow}{Discount\ rate}$$

For example, if a property generates 100,000 SAR in annual rental income and the discount rate is 8%, the present value of the perpetuity is 1.25 million SAR. This concept is valuable for evaluating the long-term profitability of real estate investments.

Importance of Present Value

Evaluates Investments

Evaluates investment opportunities.

Compares Cash Flows

Helps in comparing different cash flows.

Informed Decisions

Essential for making informed financial decisions.

Conclusion

1. **Financial Decision-Making**
   Essential for financial decision-making.

2. **Foundation**
   Foundation for advanced financial concepts.

3. **Sound Investments**
   Key to making sound investment decisions.

Finding the Return on an Investment

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Principles of Finance

Find the Return on an Investment. **– Chapter 2**

Calculating the return on an investment is a fundamental skill in finance. It provides insights into investment performance and supports informed decision-making. This presentation will cover the principles and methods for finding the return on an investment.

Learning Objectives

1. **Find the return on an investment.**

   This is the key learning objective for this section of the course. By the end, you will be able to calculate the return on any investment using the appropriate formulas and methods.

Return on Investment (ROI)

Measures gain or loss

The return on investment (ROI) measures the gain or loss generated by an investment relative to the amount of money initially invested.

Expressed as a percentage

ROI is typically expressed as a percentage, making it easy to compare the performance of different investments or benchmark against a target rate of return.

Calculating the Return

1

Formula

$$r = (FV/PV)^{(1/t)} - 1$$

2

Example

$500,000 growing to $750,000 in 4 years

= 10.67% annual return

Excel

To calculate the annual rate of return, simply type

=RATE(4, 0, -500000, 750000)

into an Excel cell and **press Enter.**

The initial investment was $500,000, and the final value grew to $750,000 over 4 years. Excel will show you the annualized percentage return.

Financial Calculator

-500,000	750,000
Present Value	Future Value
4	=10.67%
Years	Annual Return

To calculate the annual rate of return, we enter the present value (-$500,000), future value ($750,000), and number of periods (4 years) into a financial calculator. The result is an annualized return of 10.67%.

Importance of ROI

Evaluates investment performance

ROI provides a quantitative measure to evaluate the performance of an investment over a given period.

Enables comparison

By expressing returns as a percentage, ROI allows for an apples-to-apples comparison of different investment opportunities.

Informs resource allocation

Understanding the ROI of various investments can inform strategic decisions about how to allocate resources most effectively.

Conclusion

1 Fundamental skill in finance

Calculating the return on an investment is a fundamental skill that every finance professional must master.

2 Provides insights into investment performance

ROI analysis provides valuable insights into the performance of investments over time, enabling data-driven decision-making.

3 Supports informed decision-making

By understanding and applying ROI calculations, finance professionals can make more informed decisions about where to allocate capital for maximum returns.

Key Takeaways

ROI Calculation

Master the formula for calculating the return on an investment, using the future value, present value, and time period.

Evaluate Performance

Use ROI analysis to evaluate the performance of investments and compare different opportunities.

Inform Decisions

Apply ROI insights to inform strategic decisions about resource allocation and capital investment.

Continuous Learning

Calculating returns is a fundamental finance skill that requires continuous learning and practice to refine.

Calculating Investment Duration

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Principles of Finance

Calculate How Long It Takes for an Investment to Reach a Desired Value – Chapter 2

Calculating how long an investment will take to reach a desired value is a crucial aspect of financial planning. This presentation will explore the principles and formulas involved in determining investment duration, highlighting its importance in setting and achieving realistic financial goals.

Learning Objectives

1 **Calculate Investment Duration**

Calculate how long it takes for an investment to reach a desired value.

Investment Duration Concept

Definition

Time required for an investment to grow to a specified future value.

Importance

Essential for financial planning and goal setting.

Calculating Investment Duration

Formula

$$FV = PV \times (1 + r)^t$$

Rearranged

$$(FV/PV) = (1 + r)^t$$

Example

10,000 doubling at 8% interest

= 9 years

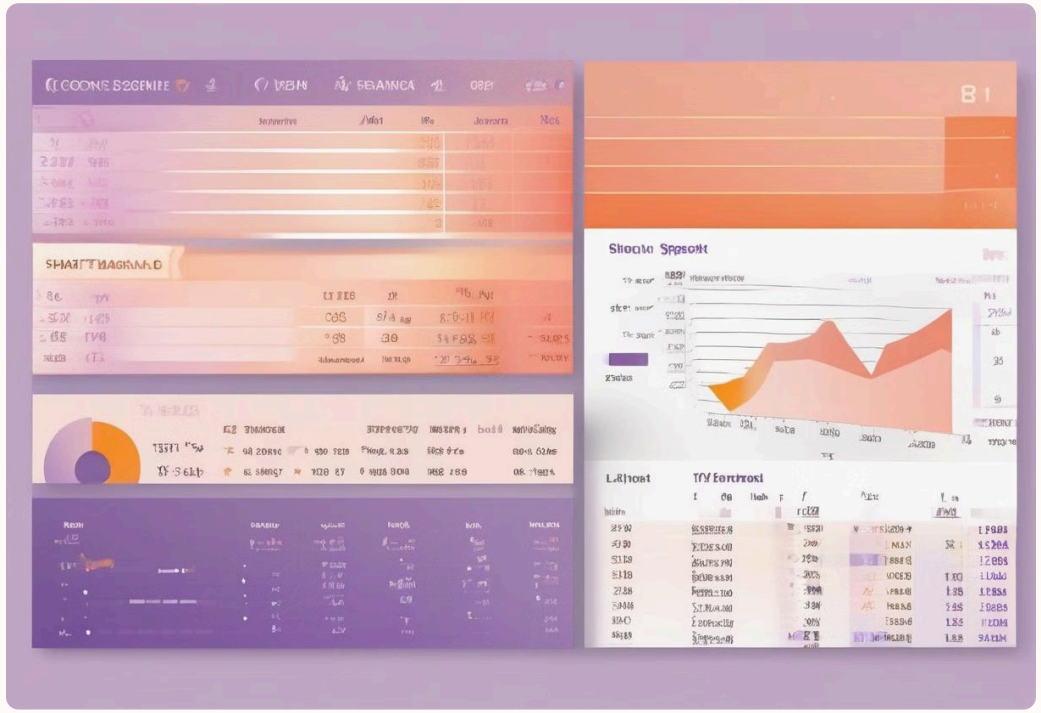
Excel

Investment Duration Using Excel

Type the following formula

```
=NPER(0.08, 0, -10000, 20000)
```

and **press Enter:**



Investment Duration Using a Financial Calculator

Enter Present Value (PV)	Enter Future Value (FV)	Enter Interest Rate (I/Y)	**Compute** Number of Periods (N)
-10,000	20,000	8%	9

Importance of Investment Duration

Financial Planning

Supports effective financial planning.

Goal Setting

Helps set and achieve realistic financial goals.

Timeline

Provides a clear timeline for financial objectives.

Conclusion

Calculating Investment Duration is:

A fundamental skill in finance

Supports realistic goal setting and financial planning

Essential for achieving financial objectives